
◆ STRING SEARCH & COUNT

1. Given the string **"python programming language"**, write a Python program to find the **first occurrence** of the substring **"gram"** between valid positions.
2. Given the string **"data science data analysis data"**, write a Python program to find the **last occurrence** of the substring **"data"** in the string.
3. Given the string **"welcome to python world"**, write a Python program to search whether the substring **"python"** is present in the string.
4. Given the string **"java python java c java"**, write a Python program to find the **last position** of the substring **"java"**.
5. Given the string **"abababab"**, write a Python program to count how many times the substring **"ab"** appears in the string.

◆ CASE CONVERSION

6. Given the string **"hello world"**, write a Python program to convert the entire string into **uppercase**.
7. Given the string **"PYTHON PROGRAMMING"**, write a Python program to convert the entire string into **lowercase**.
8. Given the string **"python programming language"**, write a Python program to convert the string so that the **first letter of each word is capitalized**.
9. Given the string **"welcome to python"**, write a Python program to convert only the **first character** of the string into uppercase.
10. Given the string **"PyThOn ProGraM"**, write a Python program to **change uppercase letters to lowercase and lowercase letters to uppercase**.

◆ STRING VALIDATION

11. Given the string **"Python"**, write a Python program to check whether the string contains **only alphabetic characters**.
12. Given the string **"2025"**, write a Python program to check whether the string contains **only numeric characters**.
13. Given the string **"hello"**, write a Python program to check whether all characters in the string are **lowercase**.
14. Given the string **"WELCOME"**, write a Python program to check whether all characters in the string are **uppercase**.

15. Given the string **"Python123"**, write a Python program to check whether the string contains **both letters and digits only**.
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◆ **STRING MATCHING**

16. Given the string **"python programming"**, write a Python program to check whether the string **starts with** the word **"python"**.
17. Given the string **"file_upload.pdf"**, write a Python program to check whether the string **ends with** the extension **".pdf"**.
18. Given the string **"learning python is fun"**, write a Python program to check whether the word **"python"** is present anywhere in the string.
19. Given the strings **"HelloWorld"** and **"helloworld"**, write a Python program to compare the two strings **ignoring case sensitivity**.
20. Given the strings **"admin123"** and **"admin123"**, write a Python program to check whether both strings are **exactly equal**.